Heejun **Park**

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"Failure is but a stepping stone to success."

Education

KAIST (Korea Advanced Institute of Science and Technology) *Daejeon, South Korea*

M.S. in Division of Future Vehicle *Sep. 2024 - Expected Aug. 2026*

- Thesis: RAF: Reliability-Aware Fusion of Camera, LiDAR and 4D Radar for Robust 3D Object Detection in Adverse Weather
- Advisor: Kuk-Jin Yoon
- GPA: 3.83 / 4.3

Yonsei University *Wonju, South Korea*

B.S. in Computer Engineering *Feb. 2018 - Aug. 2024*

- GPA: 4.24 / 4.3
- Graduated Summa Cum Laude

Technical Skills

Programming Python, C, C++, Bash
Libraries PyTorch, Open3D, OpenCV
Expertise Multi-Sensor 3D Object Detection, VLM, E2E Driving

Research & Project Experience

KAIST, Visual Intelligence Lab (Prof. Kuk-Jin Yoon) *Daejeon, South Korea*

Master's Student *Sep. 2024 - PRESENT*

- Developed a camera-LiDAR-4D radar fusion method that accounts for camera uncertainty under adverse weather conditions.
- Designed a fusion strategy for integrating infrastructure features and vehicle features in vehicle-to-infrastructure (V2I) cooperative perception.
- Developed a world model-enhanced E2E framework for cross-domain generalization in autonomous driving, in collaboration with 42dot.
- Built an HD map from vehicle-collected LiDAR data and trained an end-to-end model for autonomous driving.

Sungkyunkwan University, AI & Media Lab (Prof. Sungeun Hong) *Seoul, South Korea*

Undergraduate Researcher *Jan. 2024 - Jul. 2024*

- Developed a multi-frame fusion pipeline to generate a fused SemanticKITTI dataset for scene completion tasks.
- Built a pipeline converting NYU Depth v2 into point cloud data for 3D semantic segmentation training.
- Conducted an empirical bag-of-tricks study on ResNet-50 and ViT classifiers, achieving up to 10% accuracy improvement on CIFAR-10.

Yonsei University, AI Security Lab (Prof. Seok-Hwan Choi) *Wonju, South Korea*

Undergraduate Researcher *Jan. 2023 - Dec. 2023*

- Investigated the vulnerability of 3D classification models to backdoor attacks to explore AI security weaknesses.
- Built a CNN from scratch using only NumPy and trained it on MNIST, achieving 83.1% accuracy.

Yonsei University, Project Immersion Course *Wonju, South Korea*

Team Leader *Jan. 2023 - Jun. 2023*

- Designed an autonomous driving framework for a remote-controlled taxi integrating perception, localization, planning, and control.
- Implemented lane-keeping using OpenCV with Canny edge detection to detect lane edges and follow lane centers.
- Deployed a pre-trained YOLOv5 model for pedestrian detection, enabling the vehicle to stop when a pedestrian is detected within a set distance.
- Developed an odometry-based localization system with drift correction

Purdue University, IITP K-SW Square Program *Indiana, United States*

Team Leader at Global Internship Program *Sep. 2022 - Dec. 2022*

- Led a cross-institutional team of 7 students across Korea and Purdue University.
- Developed a hybrid machine learning pipeline for non-destructive sugar content measurement using image data.
- Built and publicly released a dataset of 10,500 image–Brix level pairs for reproducible research.

KAIST, Personal Project *Daejeon, South Korea*

Individual Researcher *Jan. 2026 - Present*

- Initiated **VLM-based E2E Autonomous Driving: Reasoning Strategies and Action Heads,** a project built on InternVL2 and the DriveLM-nuScenes dataset.
- Designed a parameter-efficient fine-tuning pipeline using Vision Language Adapter and LoRA.
- Engineered three distinct chain-of-thought prompting strategies.
- Implemented and compared three action head architectures (MLP, Diffusion-based, and RT-2-style tokenization).
- Released a technical report and open-source codebase targeting VLM-based E2E autonomous driving beginners.

Publication

RAF: Reliability-Aware Fusion of Camera, LiDAR, and 4D RADAR for Robust 3D Objet Detection in Adverse Weather

Heejun Park, Jaeseok Jeong, Kuk-Jin Yoon

European Conference on Computer Vision (ECCV), 2026, (Under Review)

VIPS: Vehicle-Infrastructure Cooperative Planning Benchmark via Pseudo-Simulation

Hoonhee Cho*, Giwon Lee*, Jae-Young Kang*, Hyemin Yang*, **Heejun Park***, Kuk-Jin Yoon

European Conference on Computer Vision (ECCV), 2026, (Under review)

HEAT: Heterogeneous End-to-End Autonomous Driving via Trajectory-Guided World Models

Hoonhee Cho*, Giwon Lee*, Jae-Young Kang*, Hyemin Yang*, **Heejun Park***, Seokwoo Jung, Kuk-Jin Yoon

European Conference on Computer Vision (ECCV), 2026, (Under review)

Test-Time Prompt Learning for Robust LiDAR Semantic Segmentation Under Distribution Shifts

Hyeonseong Kim*, Hyun-Kurl Jang*, Yujeong Chae*, Changgyoon Oh*, **Heejun Park**, Kuk-Jin Yoon

European Conference on Computer Vision (ECCV), 2026, (Under review)

VR-Drive: Viewpoint-Robust End-to-End Driving with Feed-Forward 3D Gaussian Splatting

Hoonhee Cho*, Jae-Young Kang*, Giwon Lee*, Hyemin Yang*, **Heejun Park**, Seokwoo Jung, Kuk-Jin Yoon

Conference on Neural Information Processing Systems (NeurIPS), 2025

Doppler-Aware LiDAR-Radar Fusion for Robust 3D Detection in Adverse Weather

Yujeong Chae, **Heejun Park**, Hyeonseong Kim, Kuk-Jin Yoon,

IEEE/CVF International Conference on Computer Vision (ICCV), 2025

Investigating the Vulnerability of 3D Point Cloud Classifiers to Backdoor Attacks Leveraging Transfer Learning

Heejun Park, Taewon Jung, Hanju Lee, Seokhwan Choi

Journal of Korean Institute of Intelligent Systems, 2024

Predicting the Sweetness of Multiple Apples by Applying Apple Images on Hybrid Machine Learning: Machine Learning and Deep Learning

Seokhyeon Heo, **Heejun Park**, Hanbi Kim, Jeongho Ha, Taeyun Kim, Eric Matson

Journal of Research Institute of Advanced Technology, 2024

Extracurricular Activity

Computing Thinking Course, Yonsei University *Wonju, South Korea*

Teaching Assistant *Mar. 2023 - Jun. 2023*

- Prepared class materials independently and taught Python programming to students.
- Taught computational thinking principles and their applications in problem-solving.
- Introduced students to effective use of AI-assisted tools for learning support and productivity.

Tennis Course, Yonsei University *Wonju, South Korea*

Teaching Assistant *Mar. 2022 - Jun. 2022*

- Taught students tennis techniques through step-by-step instruction.
- Prepared class materials covering the rules and history of tennis.

Open Source SW Hackathon, Yonsei University *Wonju, South Korea*

Project Developer (Individual Project) *Oct. 2021*

- Developed a disease occurrence probability prediction model using open-source historical and current health examination data.
- Built an Android application that predicts future disease risks based on user-input health examination records.

Language Proficiency

Native Korean speaker, fluent in English

- Aug. 2025 - OPIc AL
- Oct. 2020 - TOEIC 975
- Oct. 2016 - TOEFL iBT 102

Awards & Scholarships

Awards

Aug. 2024	**Graduated Summa Cum Laude**, Academic Excellence	*Yonsei University*
Nov. 2023	**Best Paper Award**, KIIS Autumn Conference	*Korean Institute of Intelligent Systems*
Nov. 2023	**Excellence Award**, SW Talent Festival Outstanding Work Award	*Ministry of Science and ICT*
Nov. 2023	**Magna Cum Laude**, Academic Excellence	*Yonsei University*
Nov. 2022	**Magna Cum Laude**, Academic Excellence	*Yonsei University*
Feb. 2022	**Encouragement Award**, GDSC Winter Hackathon	*Google Developer Student Club*
Nov. 2021	**Magna Cum Laude**, Academic Excellence	*Yonsei University*
Dec. 2021	**Excellence Award**, ASC Project Competition	*Yonsei University*
Dec. 2021	**Excellence Award**, Open Source SW Hackathon	*Yonsei University*
May. 2019	**Magna Cum Laude**, Academic Excellence	*Yonsei University*
Nov. 2018	**Magna Cum Laude**, Academic Excellence	*Yonsei University*
Aug. 2016	**KAIST Presidential Award**, King of Kings Invention Contest	*Korea International Youth Olympiad*

Scholarships

Mar. 2023	**National Excellence Scholarship (Science & Engineering)**, Mar. 2023 – Aug. 2024	*Korea Scholarship Foundation*
Mar. 2023	**Huimang Scholarship**	*Yongbok Scholarship Foundation*
Feb. 2023	**Veritas Scholarship**	*Yonsei University*
Feb. 2022	**Cheongpa Excellence Scholarship**	*Cheongpa Scholarship Foundation*
Feb. 2022	**Eunhae Scholarship**	*Eunhae Scholarship Foundation*
Feb. 2022	**Veritas Scholarship**	*Yonsei University*
Nov. 2021	**National Center of Excellence in SW Scholarship**	*National Program of Excellence in SW*
Aug. 2021	**Cheongpa Scholarship**	*Cheongpa Scholarship Foundation*
Feb. 2021	**Cheongpa Scholarship**	*Cheongpa Scholarship Foundation*
Aug. 2018	**Cheongpa Outstanding Scholarship**	*Cheongpa Scholarship Foundation*
Aug. 2018	**Veritas Scholarship**	*Yonsei University*